

MULTI-AGENT RUNTIME STACK

Review scope

ORCHESTRATION LAYER

well surveyed elsewhere

Workflow frameworks

LangGraph / CrewAI

Agent orchestration

coordination / routing

Multi-agent SDKs

execution / tools

DIAGNOSTICS LAYER

retain evidence for diagnosis

Failure localization

responsible agent / step

Observability

traces / provenance

Safety

guardrails / intervention

Benchmarks

failure / reliability

retain →

SHARED RUNTIME STATE / RETENTION BUDGET

traces | KV state | environment snapshots | execution context

→ discard / reclaim

INFRASTRUCTURE LAYER

Memory

evict / prefetch

Scheduling

place / prioritize

Cache lifecycle

reuse / TTL

Power

resource states

serve within memory / latency / power budgets